4-Slide Presentation: AI-Agentic SOC Triage & Investigation Engine

Slide 1 – What is the Project?

AI-Agentic SOC Triage & Investigation Engine

Key Points

- Develop an AI-powered SOC assistant that autonomously investigates security alerts before a human analyst reviews them.
- Use an LLM with LangGraph orchestration to gather identity, SIEM, and endpoint evidence and generate an investigation report.
- Produce a complete incident timeline, MITRE ATT&CK mapping, and risk summary while keeping the analyst in control of final decisions.

Speaker Notes

This project introduces an AI-powered Security Operations Center (SOC) investigation assistant. Instead of analysts manually collecting information from multiple security tools, the AI gathers evidence, correlates the findings, and produces a complete investigation report. The goal is to reduce repetitive work while allowing analysts to focus on making security decisions.

Slide 2 – Why is this Important?

Reducing Manual Investigation Time

Key Points

- Tier 1 SOC analysts spend significant time collecting information before determining whether an alert is a real threat.
- Traditional SOAR platforms rely on fixed workflows that cannot easily adapt to new attack scenarios.
- The AI agent dynamically decides which evidence is relevant, helping analysts investigate alerts faster and more consistently.

Speaker Notes

Security teams receive thousands of alerts every day. Much of an analyst's time is spent checking user accounts, login history, endpoint activity, and previous alerts. This project automates those repetitive investigation tasks while providing transparent reasoning that analysts can verify.

Slide 3 – How Does It Work?

Example Investigation Workflow

Key Points

- The system receives a security alert and determines which data sources should be investigated.
- It queries simulated Identity, SIEM, and Endpoint Detection & Response (EDR) services to gather evidence.

- The AI combines the results into a MITRE ATT&CK timeline, generates a natural-language report, and recommends—but never executes—a response without analyst approval.

Example Scenario

Failed Login Alert

- Alert received → Multiple failed logins detected.
- AI checks user identity, previous login history, and endpoint activity.
- AI concludes whether the activity resembles credential stuffing, summarizes the findings, and recommends whether the analyst should escalate or close the case.

Speaker Notes

This slide demonstrates the investigation process. Rather than blindly following a checklist, the AI determines which evidence is necessary for each alert. It then produces a complete investigation report, allowing the analyst to review the findings instead of gathering the information manually.

Slide 4 – Key Takeaways

Project Impact

Key Points

- Demonstrates how AI can improve SOC efficiency while maintaining human oversight through approval guardrails.
- Measures success through triage accuracy, investigation speed, and transparent reasoning using labeled test datasets.
- Provides a realistic portfolio project showcasing AI agents, cybersecurity workflows, LangGraph orchestration, and secure software engineering practices.

Closing Statement

The AI-Agentic SOC Triage Engine is designed to assist—not replace—security analysts. By automating repetitive investigations and providing explainable results, it enables faster, more consistent, and more informed cybersecurity decision-making while ensuring humans remain responsible for containment actions.

Speaker Notes

In summary, this project combines artificial intelligence with cybersecurity to create an intelligent investigation assistant. It highlights practical AI engineering, secure automation, and human-in-the-loop decision making, making it an excellent demonstration of modern SOC technology.